Ministry of Higher Education and Scientific Research University of Sfax
The National School of Electronics and Telecommunications of Sfax

A modular Platform for Near-Real-Time Data Pipeline

Final Year Project Presentation in fulfillment of the requirements for the Engineering Degree in Data Engineering and Decision Systems

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Academic year 2024/2025

SUMMARY

PROJECT CONTEXT

GENERAL CONTECT

Data from various sources and differents types

Massive Data Volumes

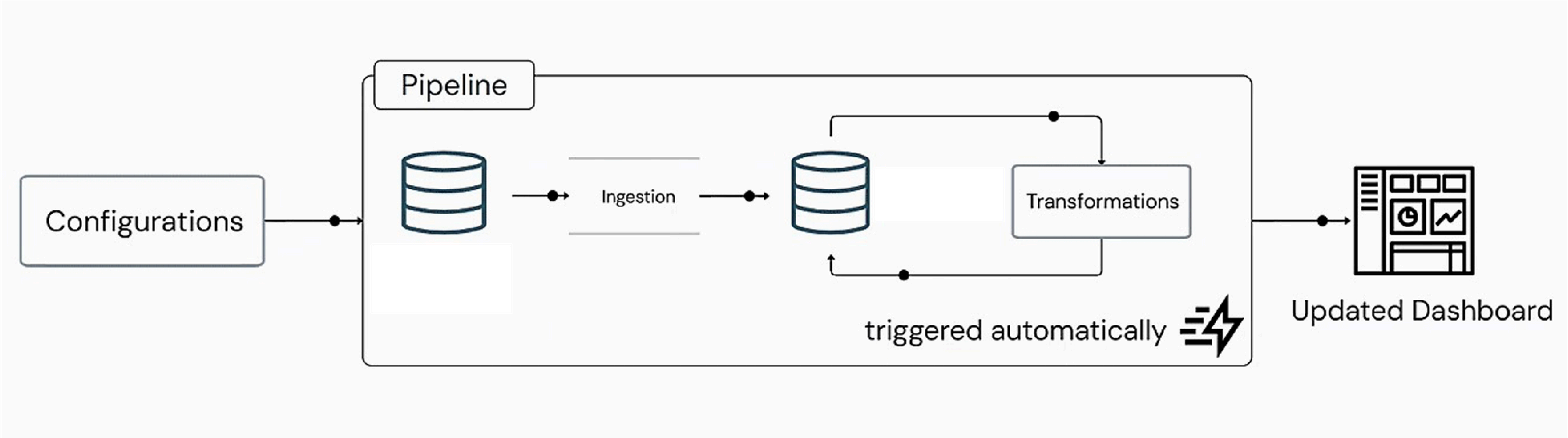
Need to make decision in near real time

CHALLENGES

- Requires technical background (Data Engineers)
- Time-Consuming Manual Tasks
- Outdated Data

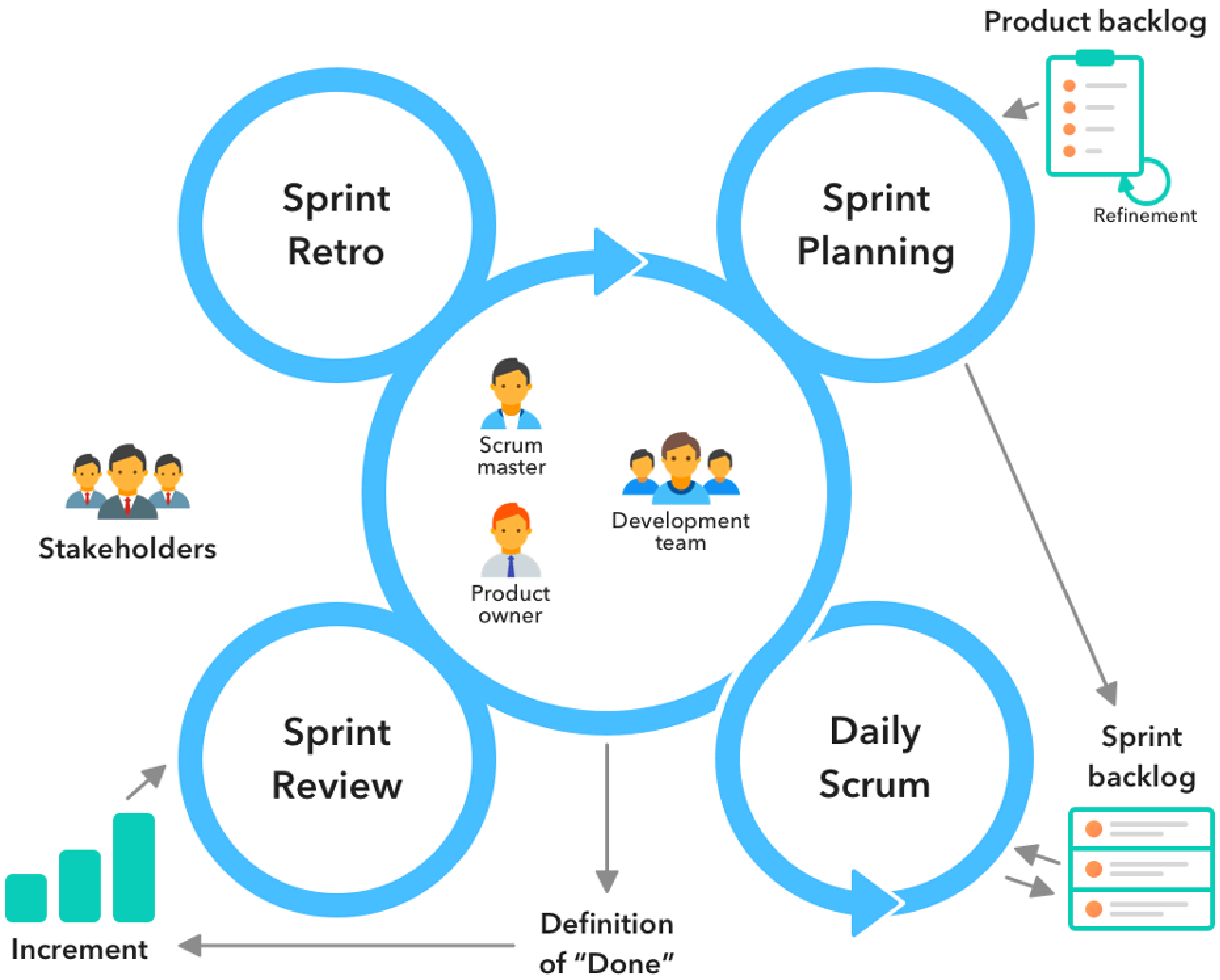
SOLUTION

💡 PROPOSED SOLUTION



Our solution is a data pipeline builder with minimal configuration that triggers the pipeline automatically as new events occur. This allows the business owner to focus on discovering insights.

💡 SOLUTIOWORK METHODOLOGY (AGILE SCRUM)



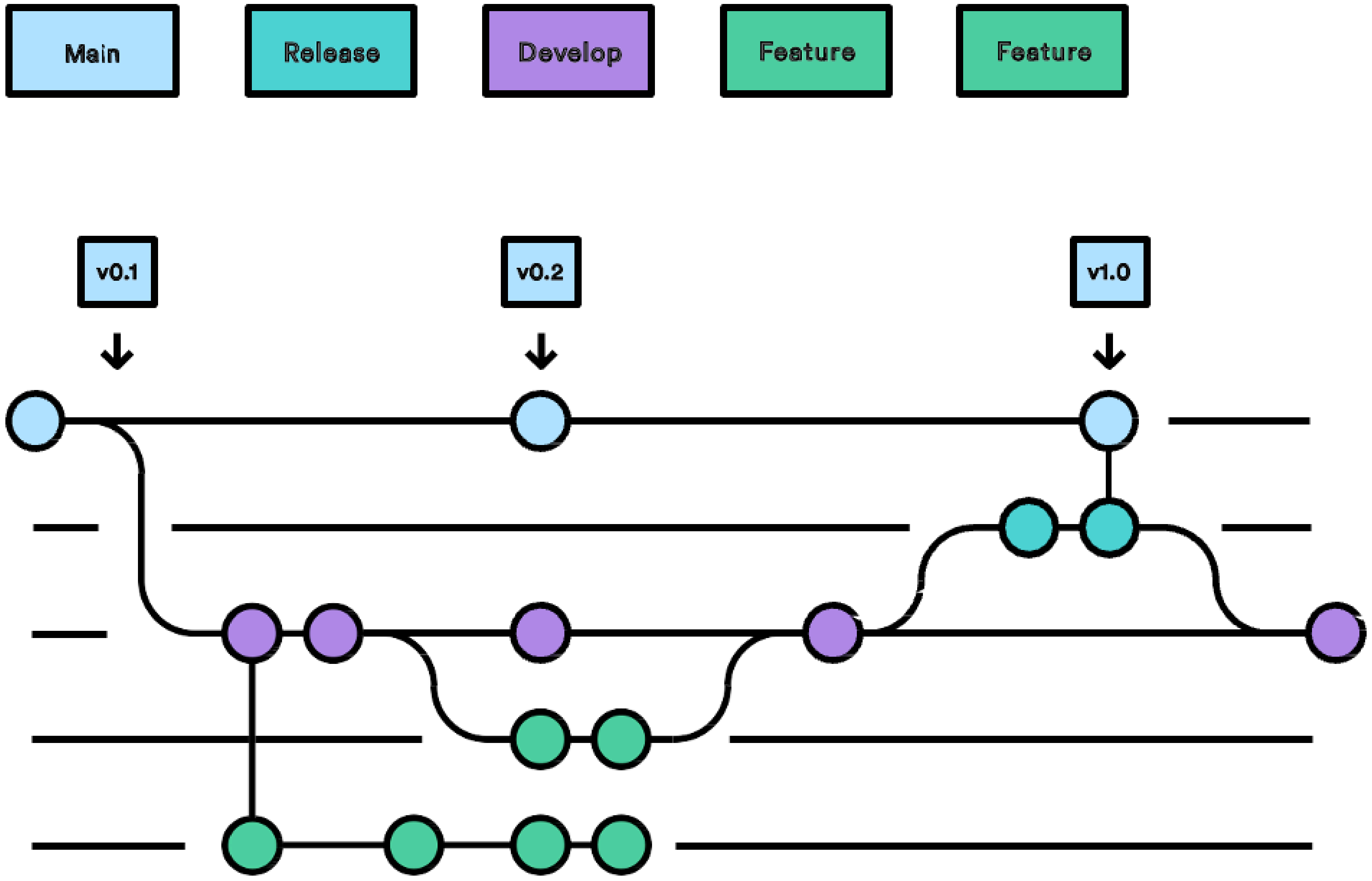
SPRINT PLANNING

Sprint 1: Data Ingestion Step	Sprint 2 :Data Transformation Step	Sprint 3:Impliment Multiple Transformation Type
• Benchmark Data Ingestion Tools • Integrate Data ingestion Tools	• Benchmark Data Transformation Tools • Integrate Data Transformation Tools	• Integrate Simple, Complex, and Agentic Transformation

SPRINT PLANNING

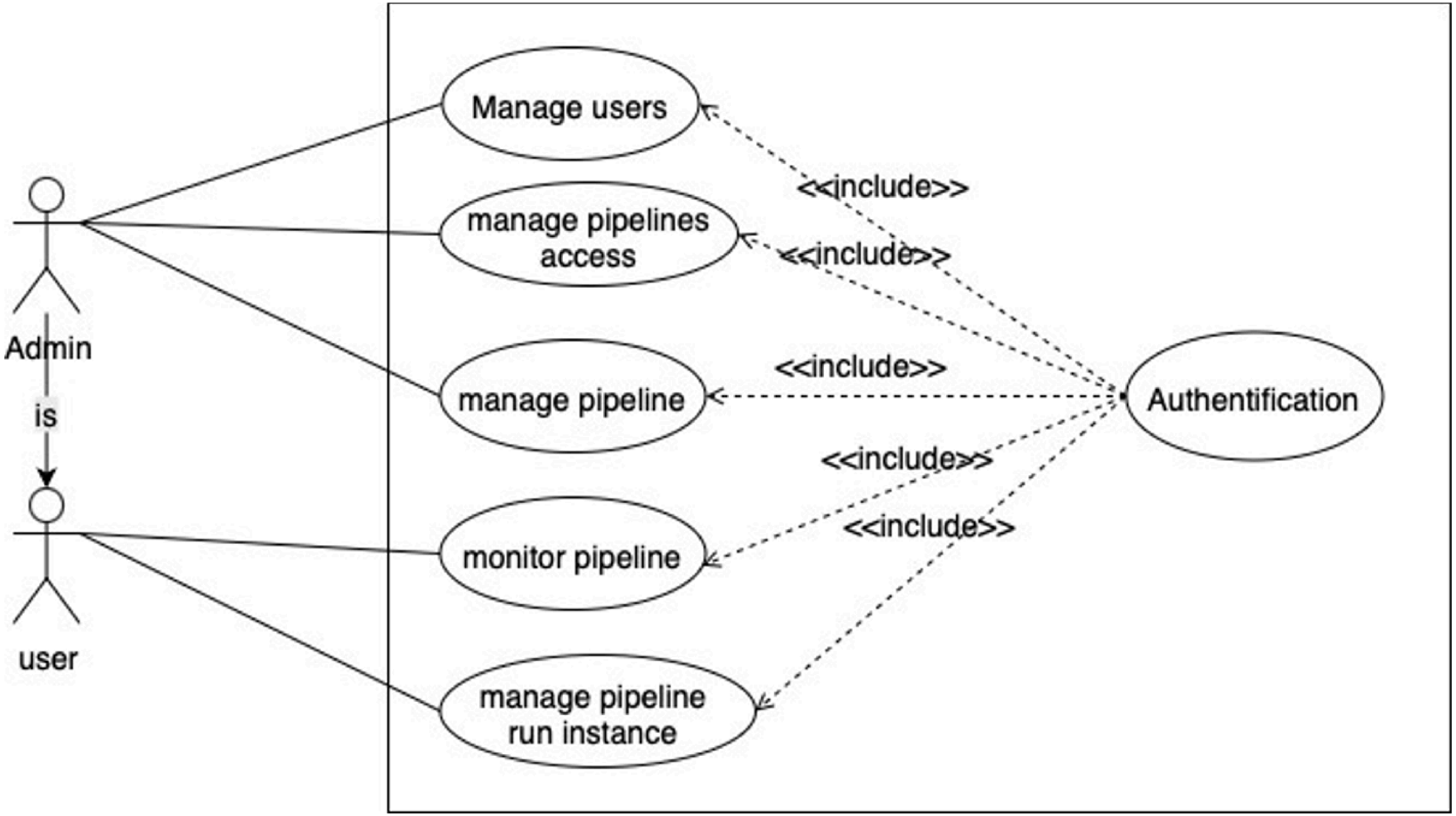
Sprint 4: Data Visualiaztion Step	Sprint 5: Role Based Access Control	Sprint 6: Schema Change Detection
• ·Integrate Data Visualization tool	• ·Integrate Role Based Access Control	• Integrate Schema Change Detection • Integrate Event Driven Schedular

DEVELOPMENT WORKFLOW (GITFLOW)

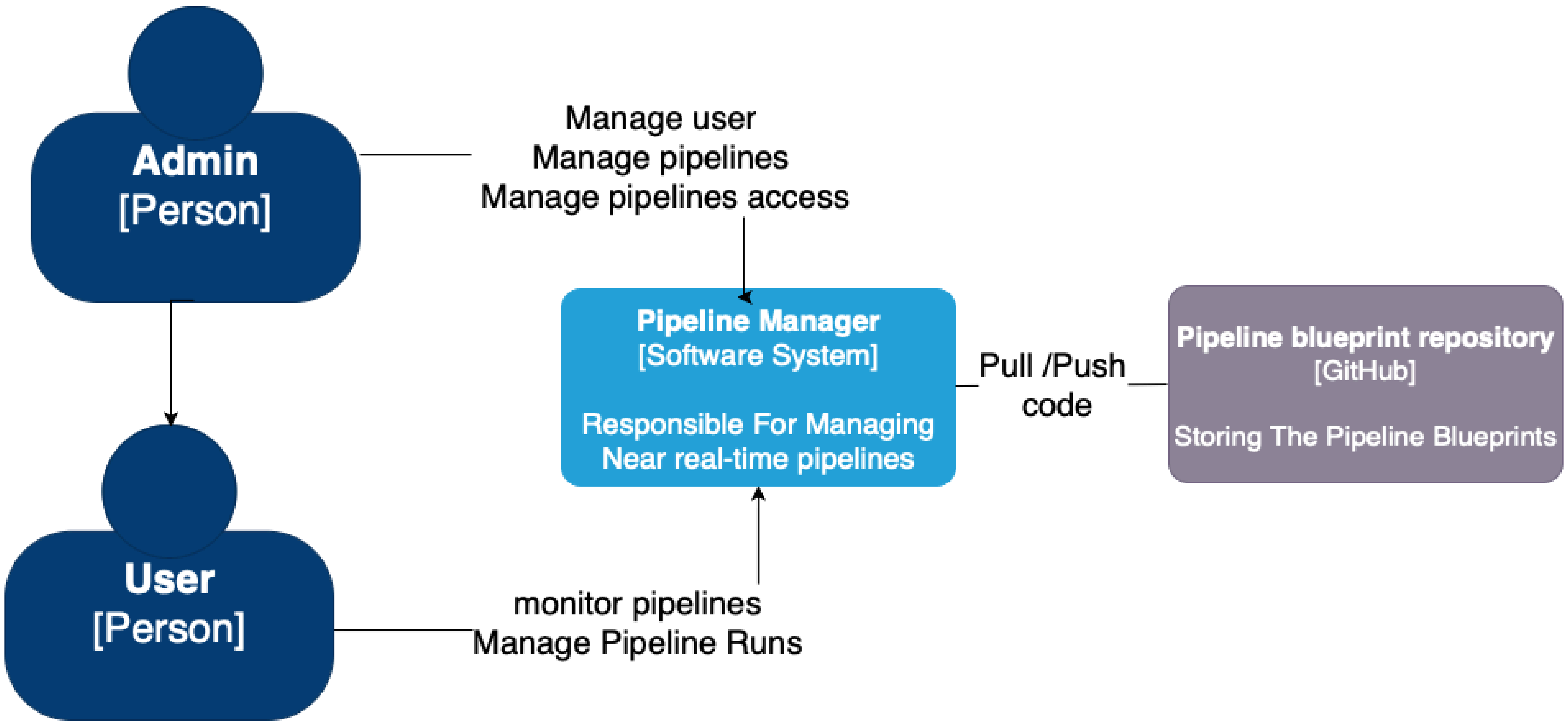


- Clear Separation of Concerns
- Parallel Development

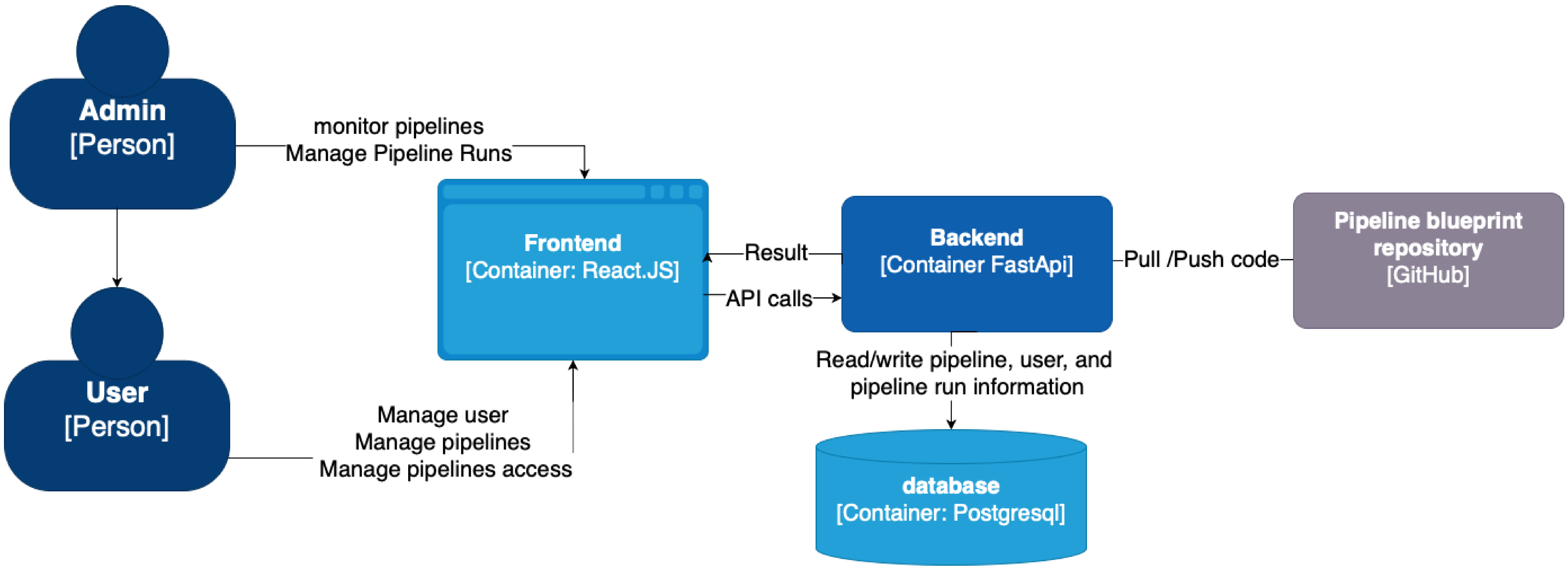
USE CASE DIAGRAM



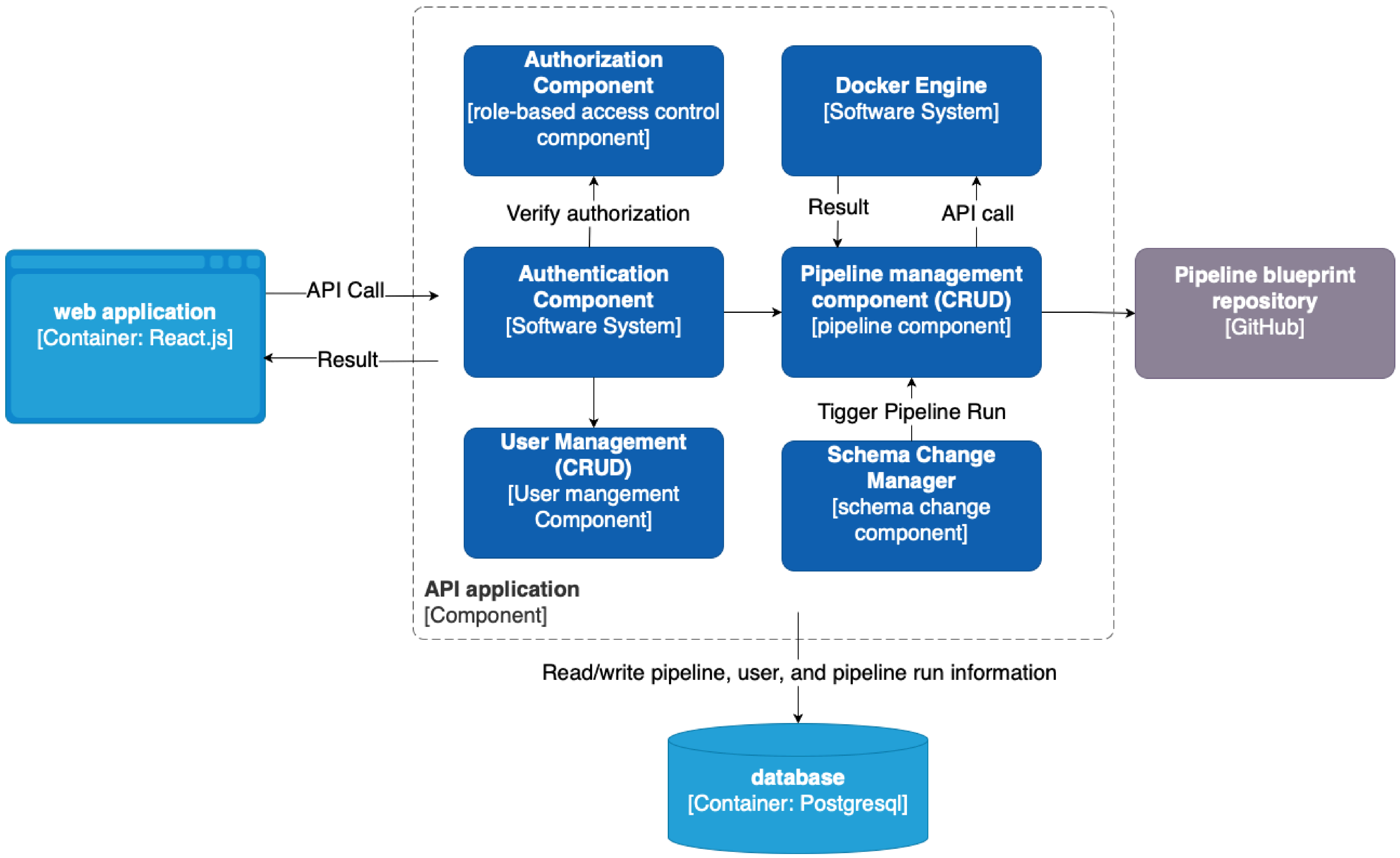
SYSTEM CONTEXT DIAGRAM



CONTAINER DIAGRAM



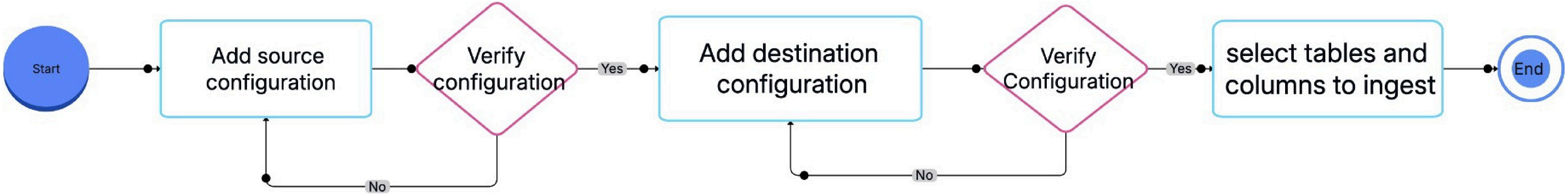
COMPONENT DIAGRAM



ACHIEVEMENT

DATA INGESTION STEP

WORKFLOW



DATA INGESTION STEP

TOOLS PRESENTATION

AIRBYTE

- open-source data integration platform
- 350+ pre-built connectors
- Custom connector creation with low-code

DATA LOAD TOOL

- Python-based framework
- Handles schema evolution and performs data validation

APACHE NIFI

- open-source platform for automating and managing data flows
- drag and drop interface

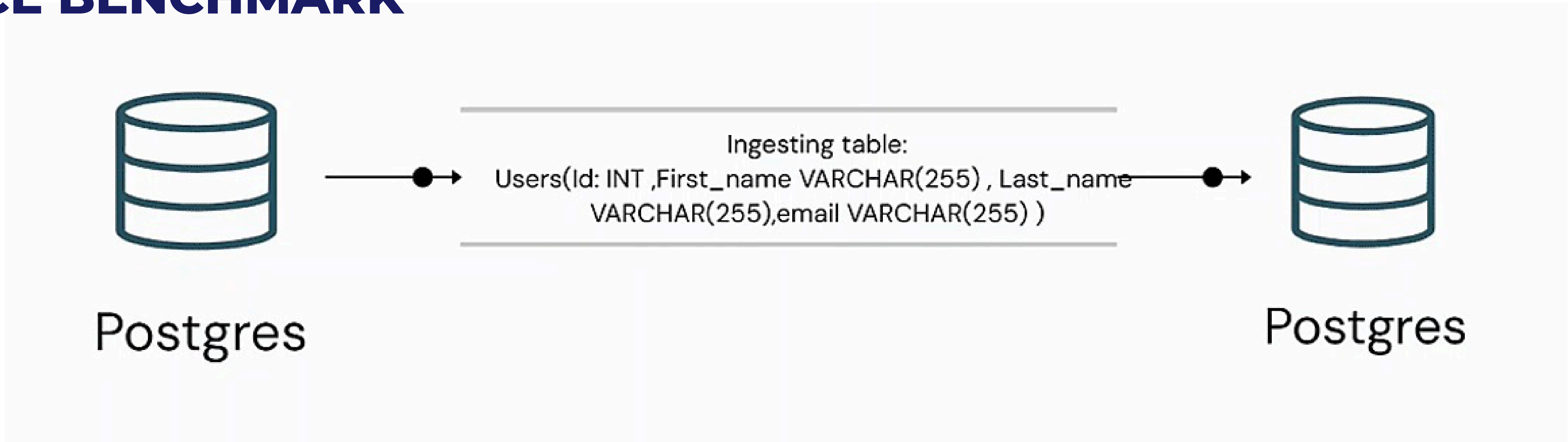
DATA INGESTION STEP

TOOLS BENCHMARK

Tools	Parallel execution Integration	Structured/Semi Structured	Low Level Code / API Integration	Schema change detection	Third-party application
DLT	✓	✓	✓	✓	✓
Airbyte		✓	✓	✓	✓
Apache Nifi	✓	✓			

DATA INGESTION STEP

PERFORMANCE BENCHMARK



Note: When using Apache NiFi, the target table must be created manually .

Tools	100 rows	1k rows	10k rows	100k rows	1m rows
Airbyte	8s	10s	12s	18s	2m 22s
DLT	1s	1.3s	1.7s	2.15s	15.6s
Apache Nifi	0.36s	0.4s	0.45s	1.7s	13s

DATA INGESTION STEP

PERFORMANCE BENCHMARK

DLT CONFIGURATION;

- **Chunksize :** 100000 rows
- **Extraction:** 5 workers
- **Normalize:** 5 workers
- **Load:** 5 workers

RESULT:

Tools	500k rows	1m rows	5m rows
DLT	5s	9.2s	44.5s
Apache Nifi	4.12s	5.29s	24.73s

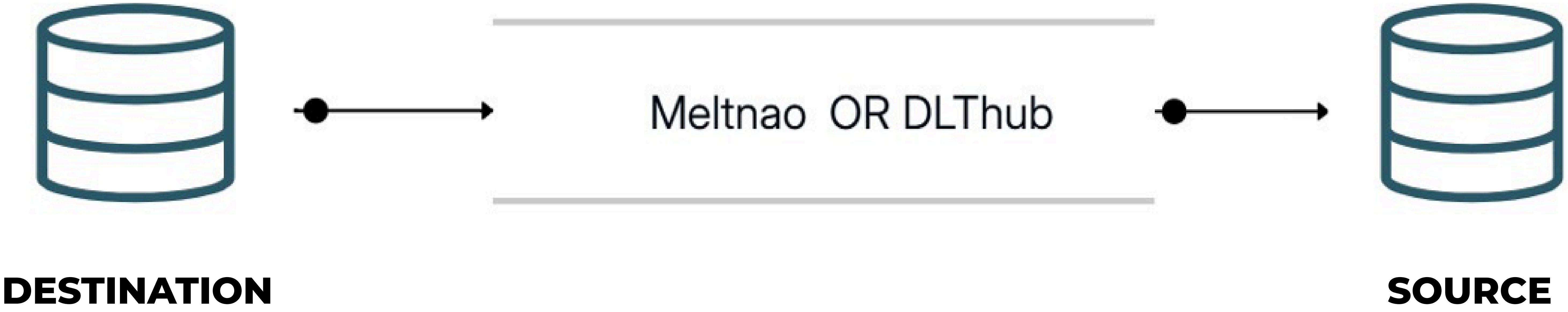
DATA INGESTION STEP

PERFORMANCE BENCHMARK

	Apache Nifi	DLT
UI Based	✓	
Code Based		✓
Integrate with Third-party applicaiton (sqlmesh,airflow ..)		✓
Detect Schema Change		✓

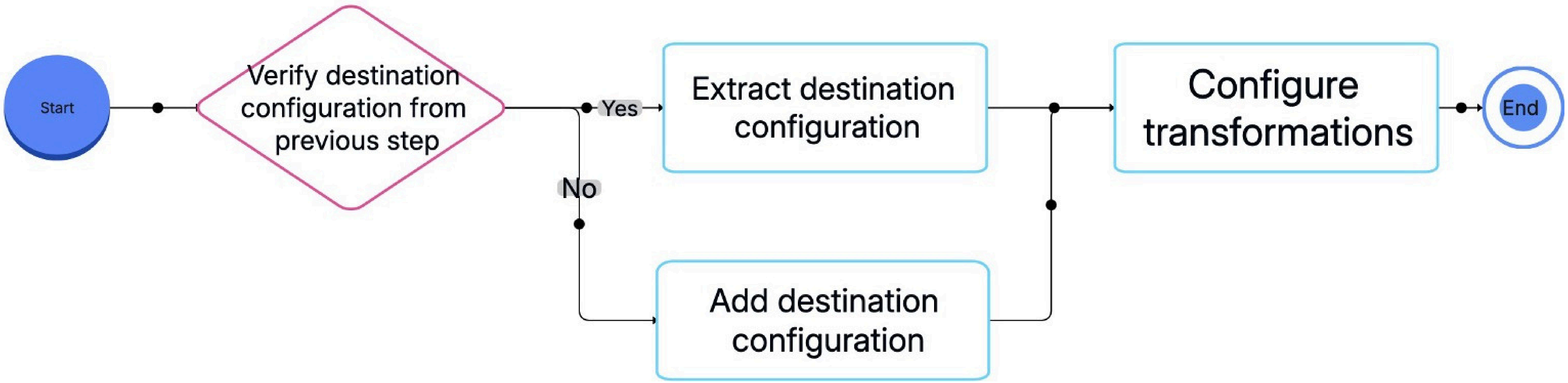
DATA INGESTION STEP

SUPPORTED TOOLS AND CONNECTORS



DATA TRANSFORMATION STEP

WORKFLOW



DATA TRANSFORMATION STEP

TOOLS PRESENTATION

DATA BUILD TOOL

- Supports SQL transformations
- Automatically runs models in the right order

SQLMESH

- Supports SQL and Python transformations
- Keeps track of what data is already processed
- Supports fast testing and development with virtual environments

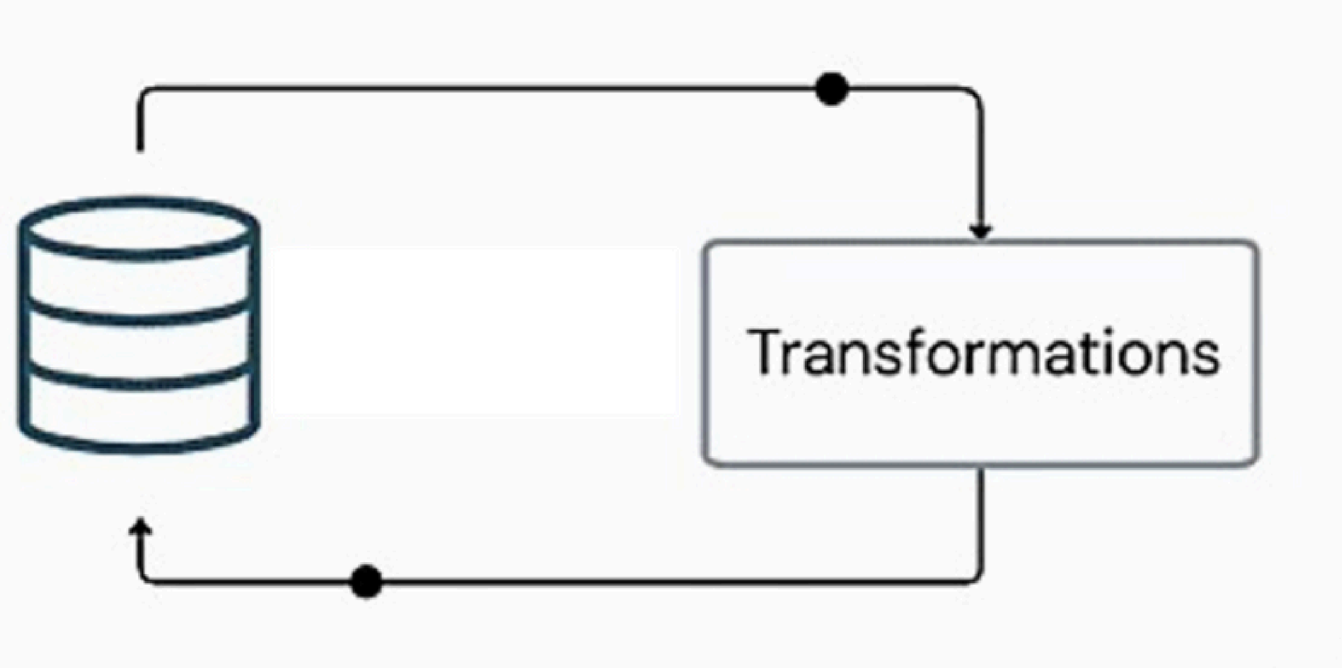
DATA TRANSFORMATION STEP

TOOLS BENCHMARK

Tools	State Manegment	Virtual Data Environments (VDEs)	Command Line Interface (CLI)	Testing & Validation
SQLmesh	✓	✓	✓	✓
DBT			✓	✓

DATA INGESTION STEP

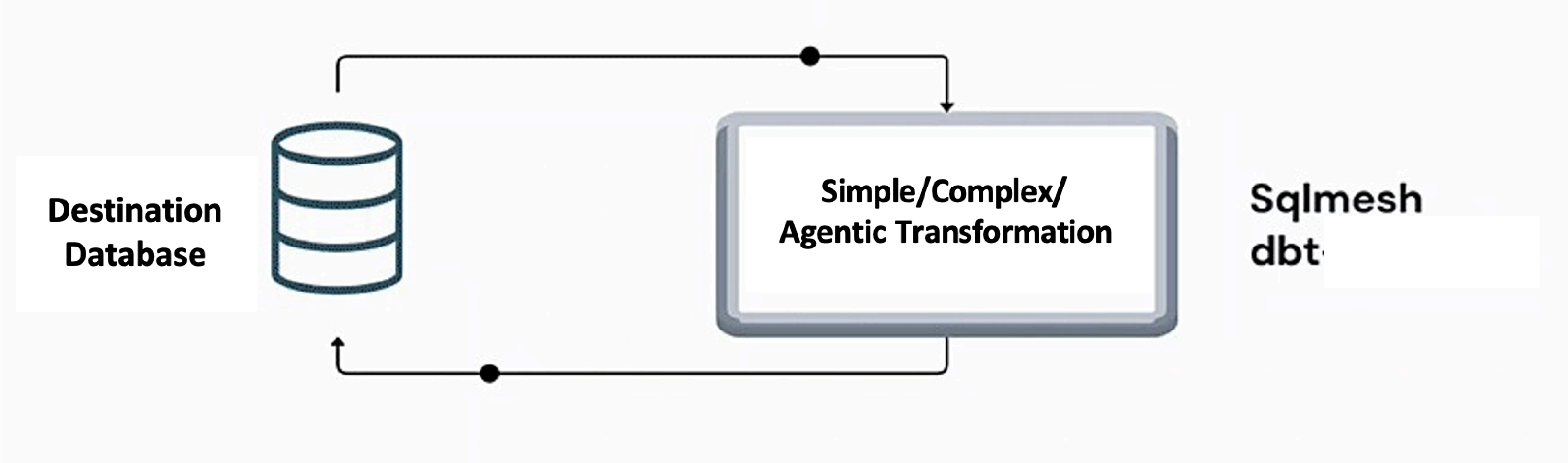
PERFORMANCE BENCHMARK



Tools	10k rows	100k rows	1m rows
DBT	12.4s	58.2s	423.7s
SQLmesh	6.8s	24.1s	146.3s

DATA TRANSFORMATION STEP

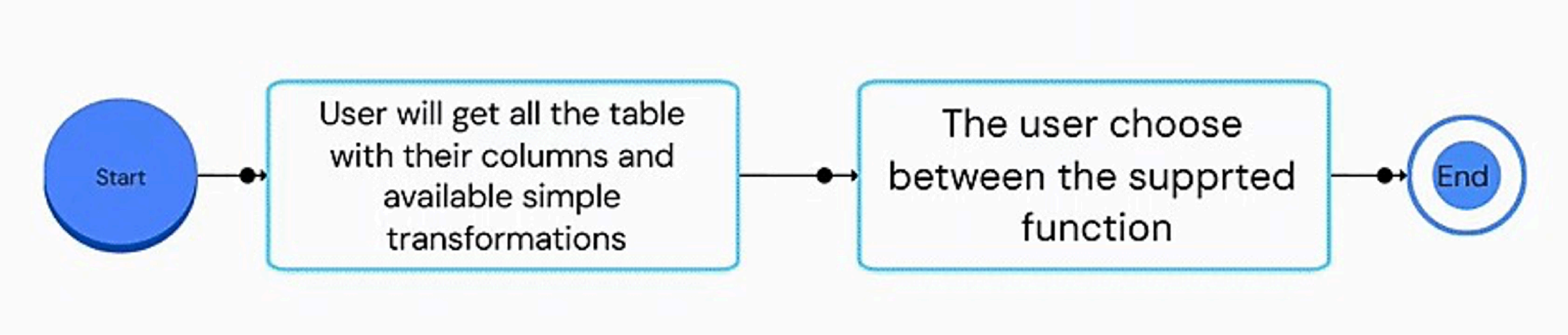
SUPPORTED TOOLS & TRANSFORMATIONS



DATA TRANSFORMATION STEP

SIMPLE TRANSFORMATIONS

- **String Tranformation:** Upper,Lower
- **Null values Imputation:**Average ,min,max and median
- **Date Transformation:** Extract day, extract year, extract month

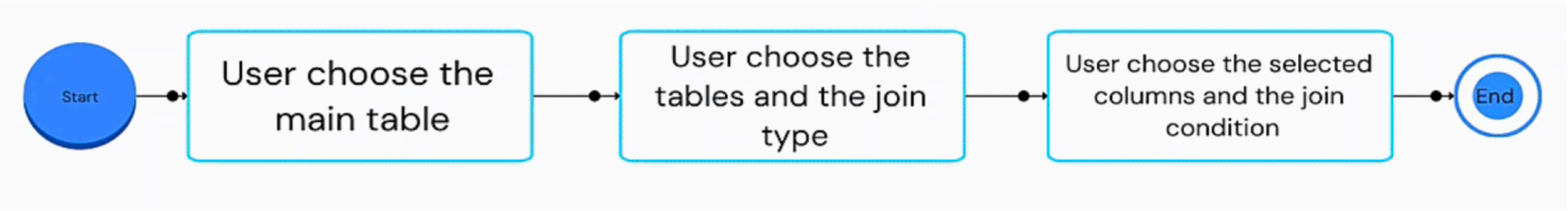


DATA TRANSFORMATION STEP

COMPLEX TRANSFORMATIONS

JOIN TRANSFORMATION

- Joining with multiple table
- Support multiple join type :Right,Left,Inner,Cross and Full Outer joins

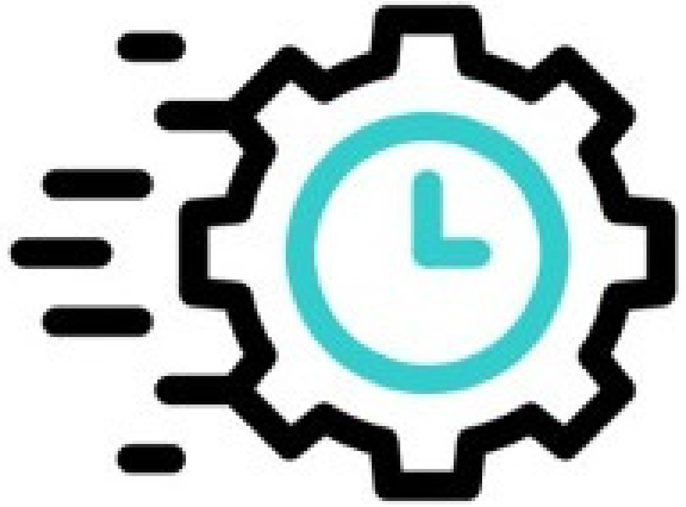


DATA TRANSFORMATION STEP

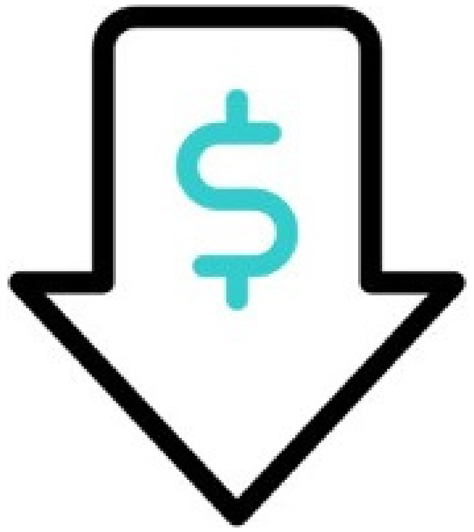
AGENTIC TRANSFORMATION

LLM CHOICE: GEMINI 2.0 FLASH

1 Million Token context Window (750 000 words).



Fast response (5seconds)



Generous free trial and Cost-effectiveness

DATA TRANSFORMATION STEP

AGENTIC TRANSFORMATION

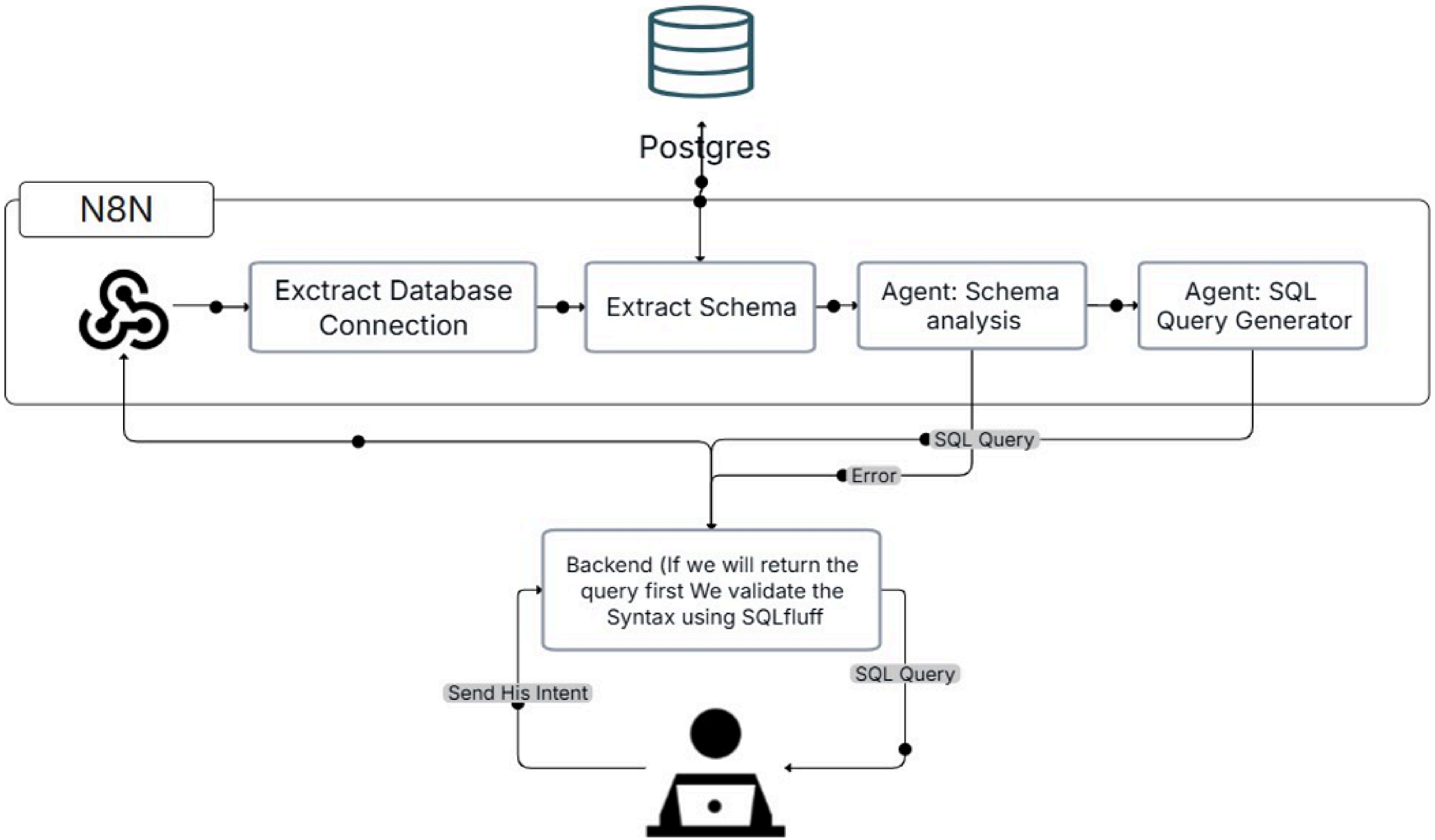
LLM INTEGRATION: N8N

End-to-End Workflow Automation

Simple drag and drop interface

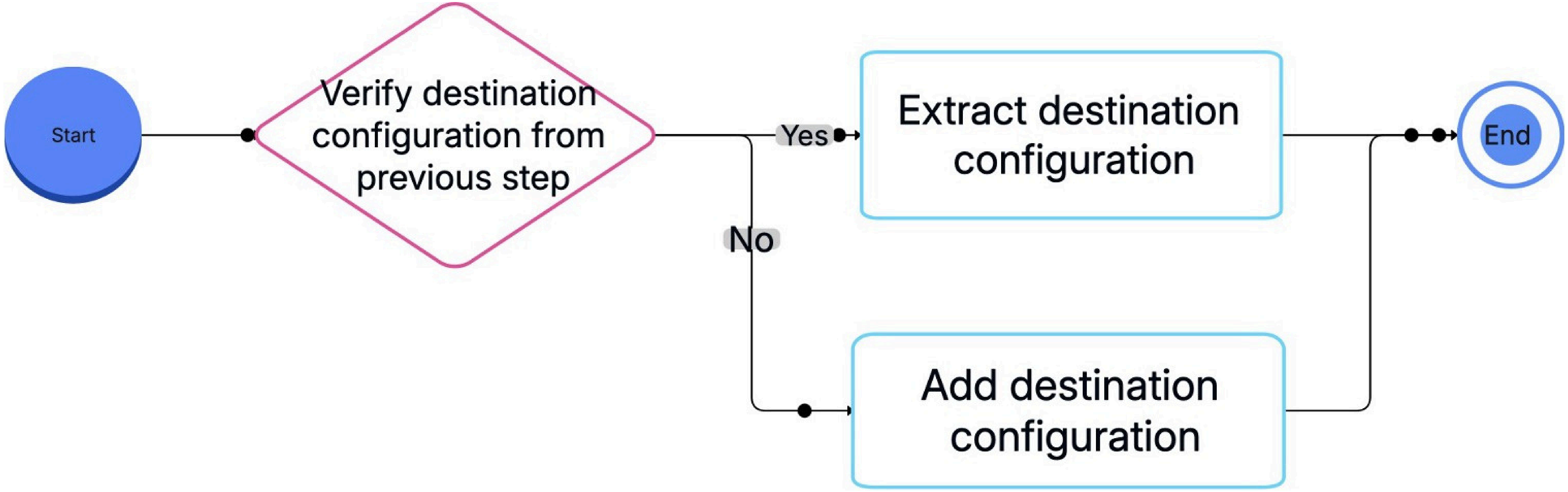
DATA TRANSFORMATION STEP

AGENTIC TRANSFORMATION WORKFLOW



DATA VISUALIZATION STEP

WORKFLOW



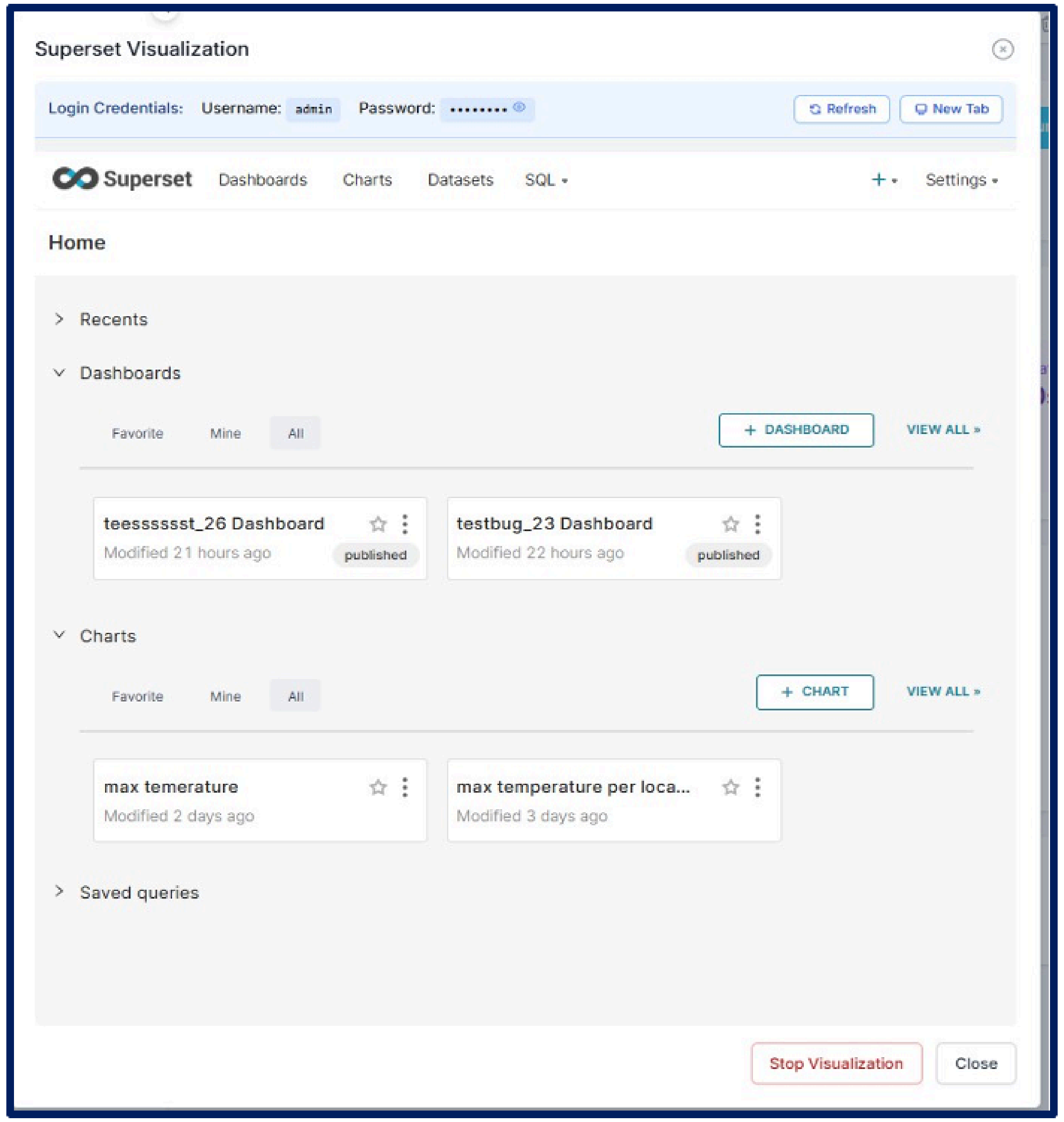
DATA VISUALIZATION STEP

TOOL PRESENTATION

WHY SUPERSET

- Open Source
- Extensive Data Source Connectivity
- Role Based Access Control

DATA VISUALIZATION STEP



After starting visualization

- Database Connection Established
- Dashboard Access Granted

ROLE BASED ACCESS CONTROL

TOOL PRESENTATION

WHY KEYCLOACK

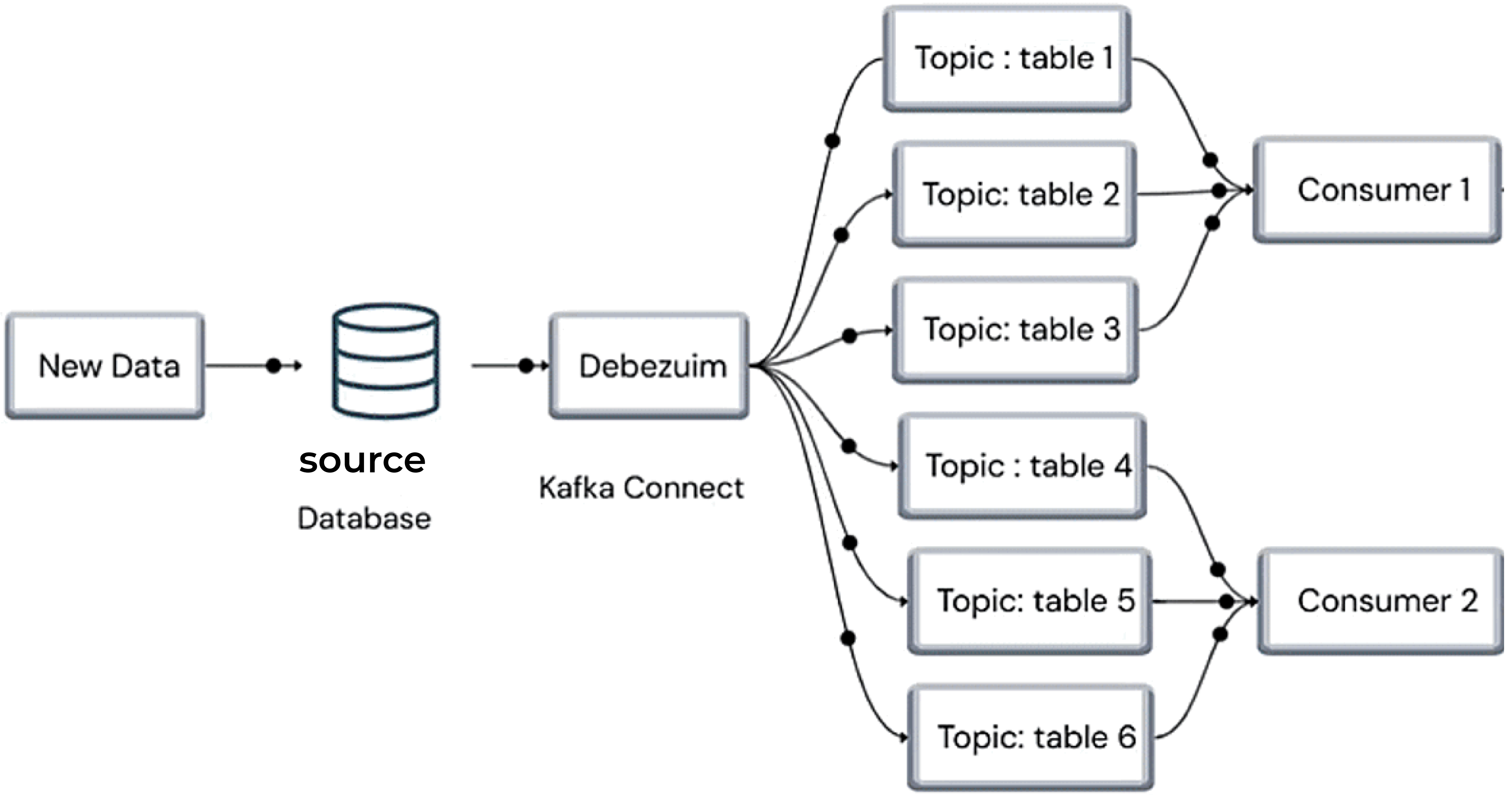
- Open Source
- Security Best Practices
- Reduced Development Overhead

ROLE BASED ACCESS CONTROL

Admin	User
Create, Delete, View, Run, and Monitor pipeline.	Monitor, and Run pipeline he have access to
Update Dashboards Superset	View Superset Dashboards only.
Manage Pipeline Access	
Manage Users	

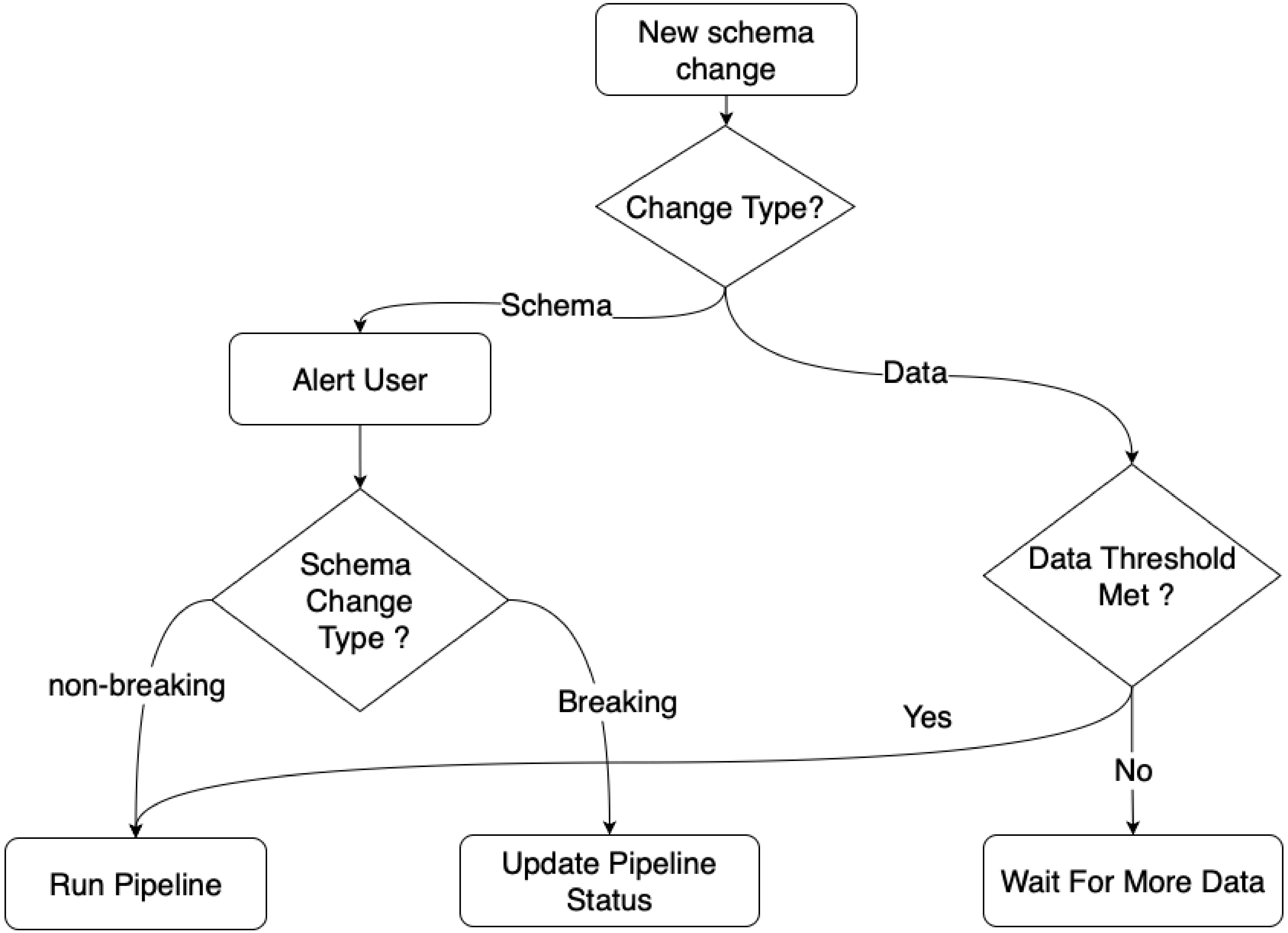
SCHEMA CHANGE DETECTION

DEBEZEUIM



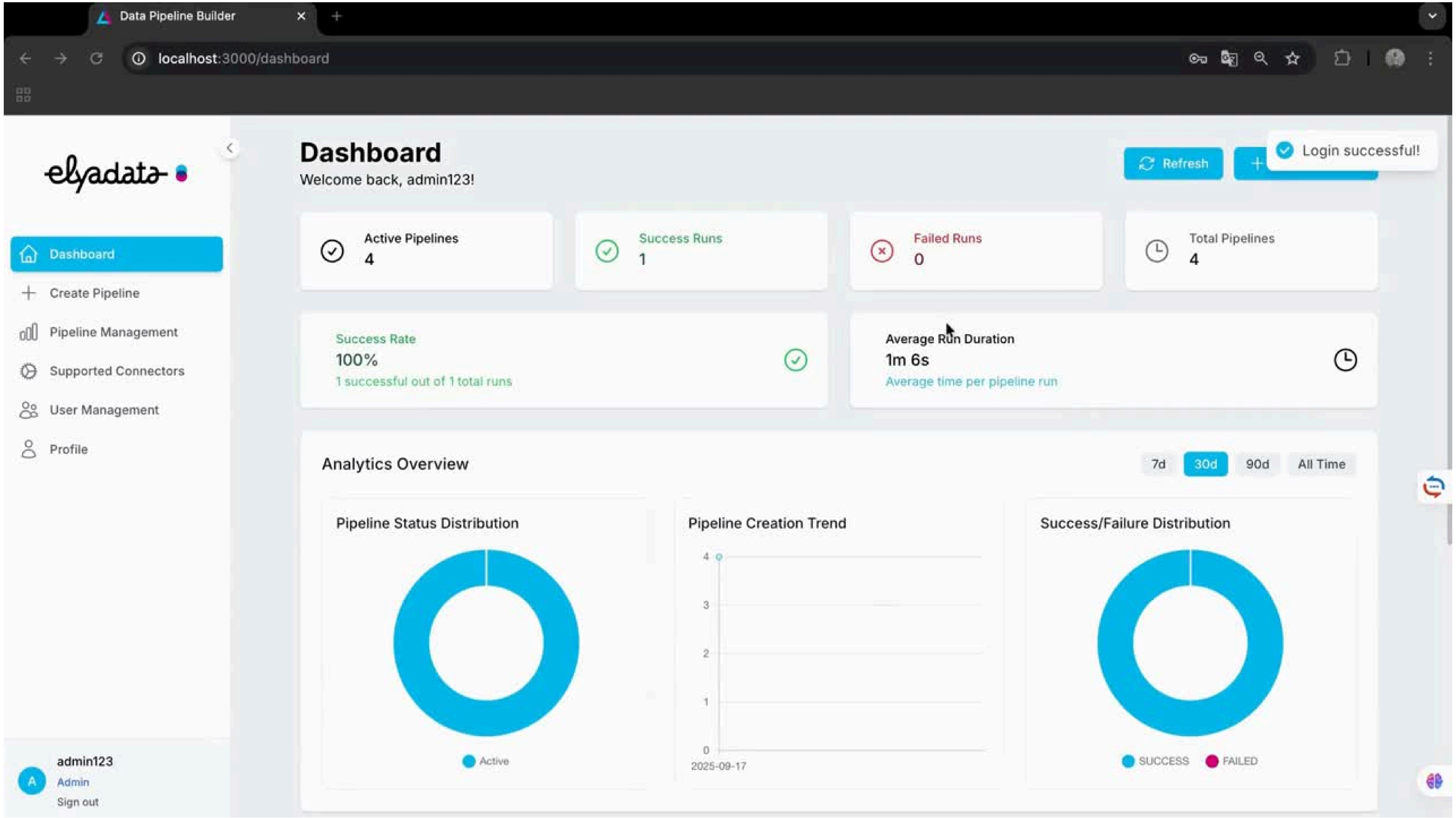
SCHEMA CHANGE DETECTION

SCHEMA CHANGE DETECTION WORKFLOW

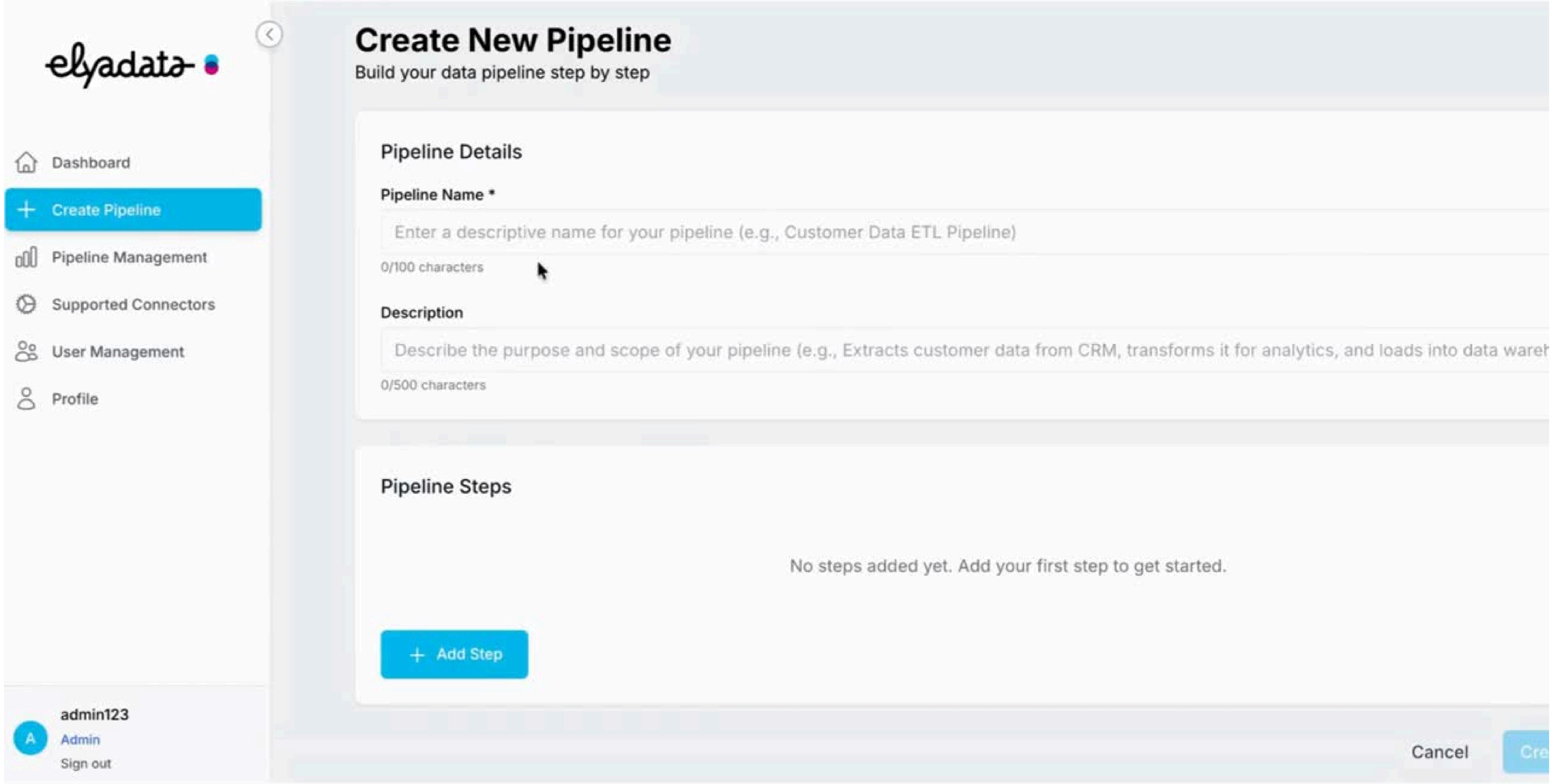


DEMONSTRATION

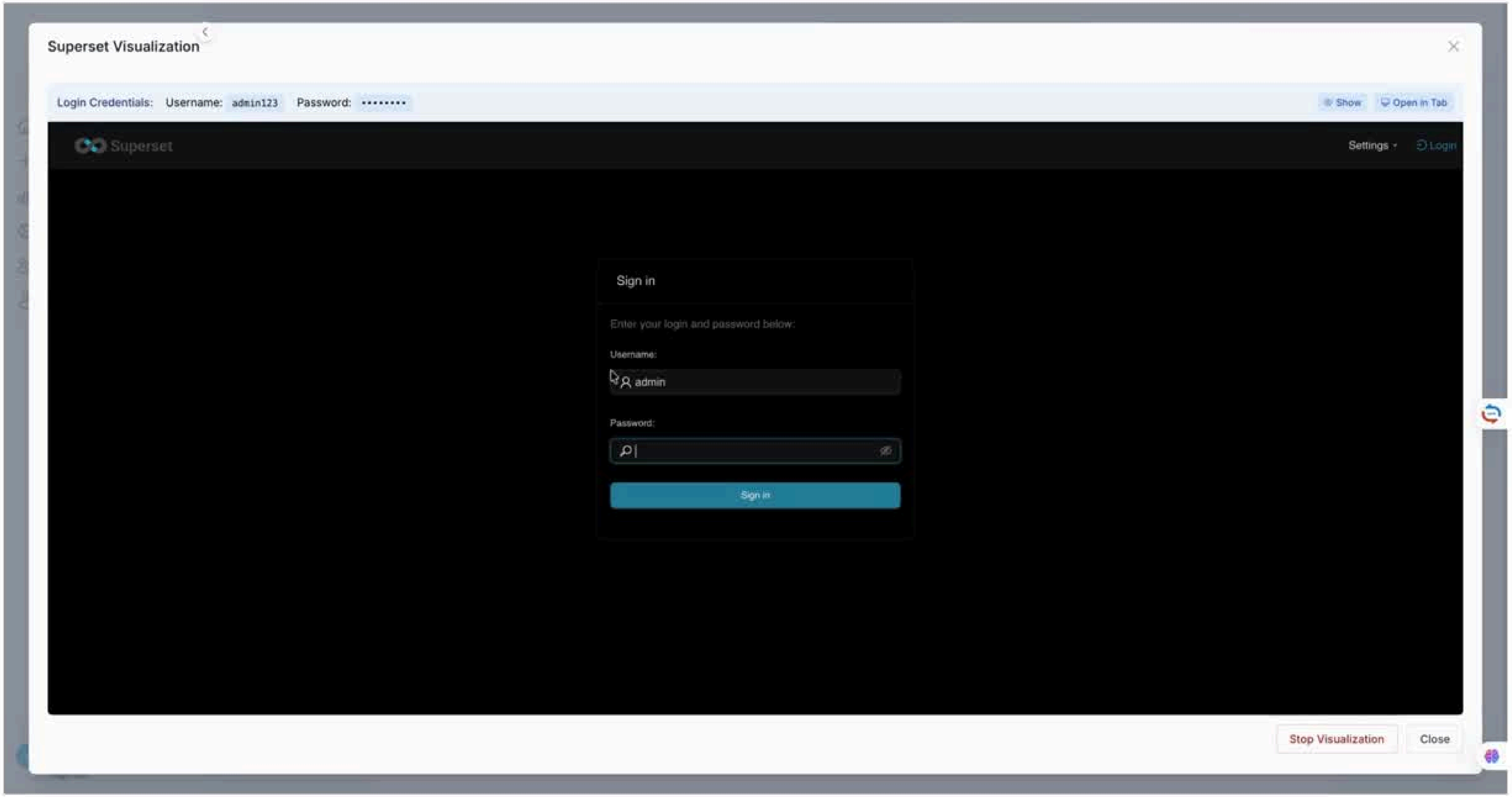
SUPPORTED CONNECTORS



CREATE PIPELINE

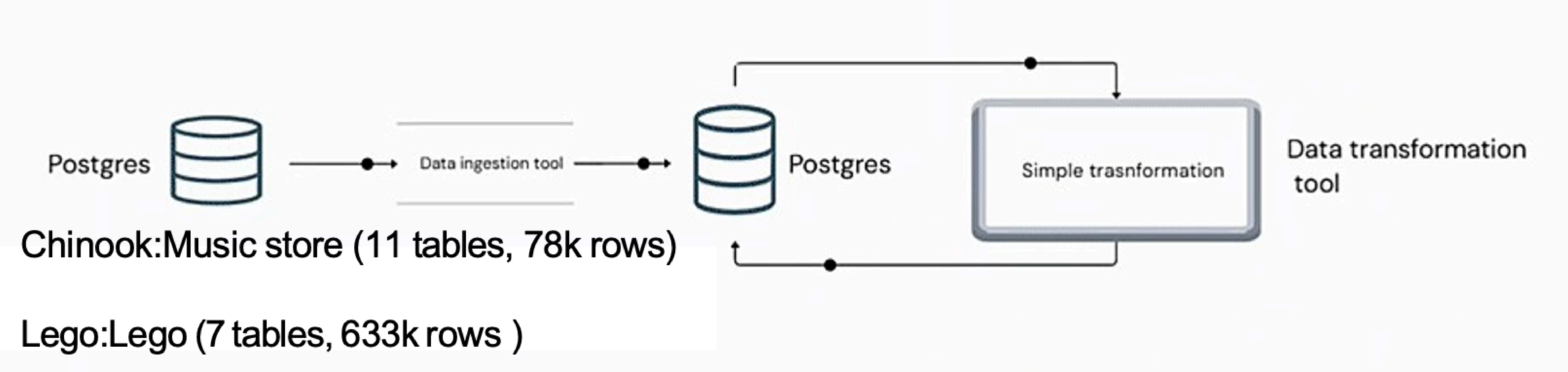


RUNNING THE PIPELINE



RESULTS AND PERSPECTIVES

RESULTS



	Chinook Database	Lego database
DLThub + SQLmesh	1min 17s	1min 41s
Meltano	3 min 40s	5min 53s

PERSPECTIVES

- Introduce data quality checks
- Enable data lineage tracking
- Integrate an agent that creates charts based on user intent and provides actionable recommendations on KPIs.

THANK YOU FOR YOUR ATTENTION

MELTANO Data lifecycle tool
Model/Extract/Load/Transform/Analyze/Notebook/Orchestrate